Electrical and Computer Engineering
University of Thessaly

ECE327 - Digital Systems VLSI

SPICE Design and Performance Analysis of CMOS Latches and Master-Slave Flip-Flops

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Contents

1 Transistor-Level Design and Simulation of a Jam Latch with Weak Inverter Feedback

The topology examined in figure 1 represents a negative-edge-triggered D-type latch implemented using transmission gates. This operation is verified by the fact that when the clock signal (CLK) is high (1), the transmission gate is non-conducting (cut-off mode). Conversely, when CLK is low (0), the input signal D is transparently propagated to the output Q."

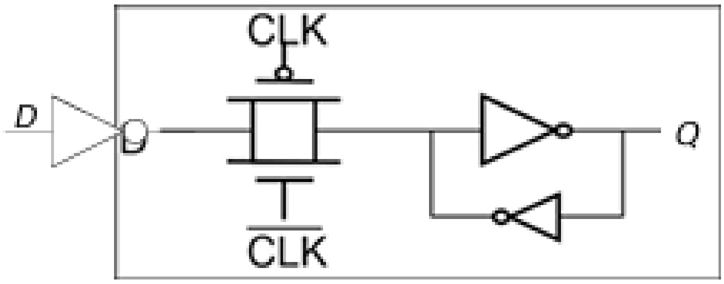

Figure 1: Latch with weak inverter feedback.

The truth table of the negative D-latch is presented below:

Table 1: Negative D-Latch Truth Table

Clock	D	Q
1	X	Q_{prev} (Previous State)
0	0	0
0	1	1

Utilizing ngspice, the latch topology was simulated to obtain the transient response waveforms(2). The output signal Q, the input data D, and the clock signal (CLK) are illustrated in the simulation plot. The resulting waveforms successfully verify the functional correctness of the circuit, in full alignment with the theoretical truth table.

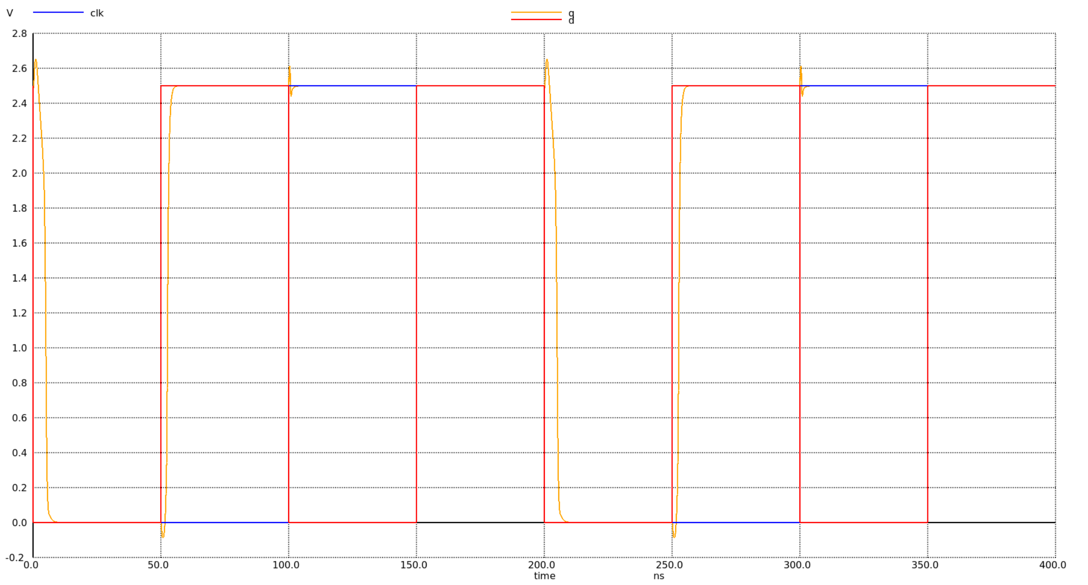

Figure 2: transient response waveforms of the topology.

2 Transistor-Level Design and Timing Characterization of a Master-Slave Flip-Flop

2.1 (a)

The topology examined in figure 3 constitutes a negative-edge-triggered Master-Slave register (Flip-Flop) designed using a multiplexer-based configuration. The internal multiplexers are implemented via transmission gates, similar to those analyzed in the previous section.When the clock signal is low ($CLK = 0$), the transmission gate T_1 is non-conducting (cut-off mode), whereas T_2 conducts, feeding its output back to the inverter and subsequently capturing the state at node Q_M.Concurrently, in the slave stage, T_3 is conducting while T_4 is in cut-off. Consequently, the output of T_3 is inverted once more to match the value of Q_M, which is then transparently mirrored at the primary output Q.

Conversely, when the clock signal switches to a high level ($CLK = 1$), T_1 becomes conducting and T_2 transitions into cut-off. Under this condition, the input data D is sampled and copied to the intermediate node Q_M. At the same time, T_3 turns off and T_4 turns on. The cross-coupled inverters of the slave latch successfully preserve its state, preventing the new value of Q_M from altering the primary output Q; instead, the previous state of the output is securely maintained.

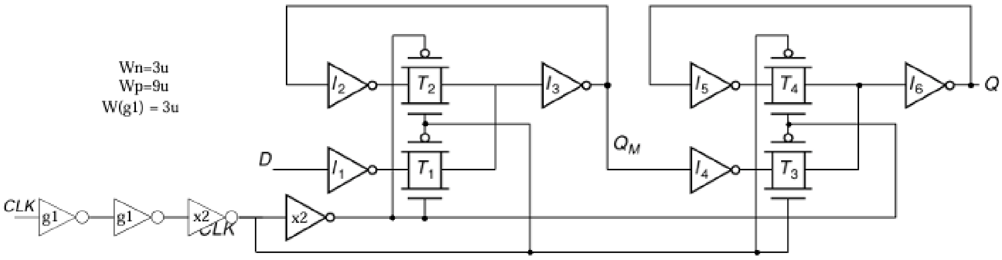

Figure 3: Master-slave register in transistors level

The truth table of the negative-edge-triggered D flip-flop is presented below:

Table 2: Negative-Edge-Triggered D Flip-Flop Truth Table

Clock	**D**	**Q(n+1)**
1 or 0 (Stable)	X	Q_n (No change)
↓ (Falling Edge)	0	0
↓ (Falling Edge)	1	1

By utilizing the ngspice simulation tool, the transient response waveforms(5)were obtained. These simulation results successfully verify the operational integrity of the Master-Slave Flip-Flop, demonstrating a perfect alignment with the theoretical truth table (Table 2).

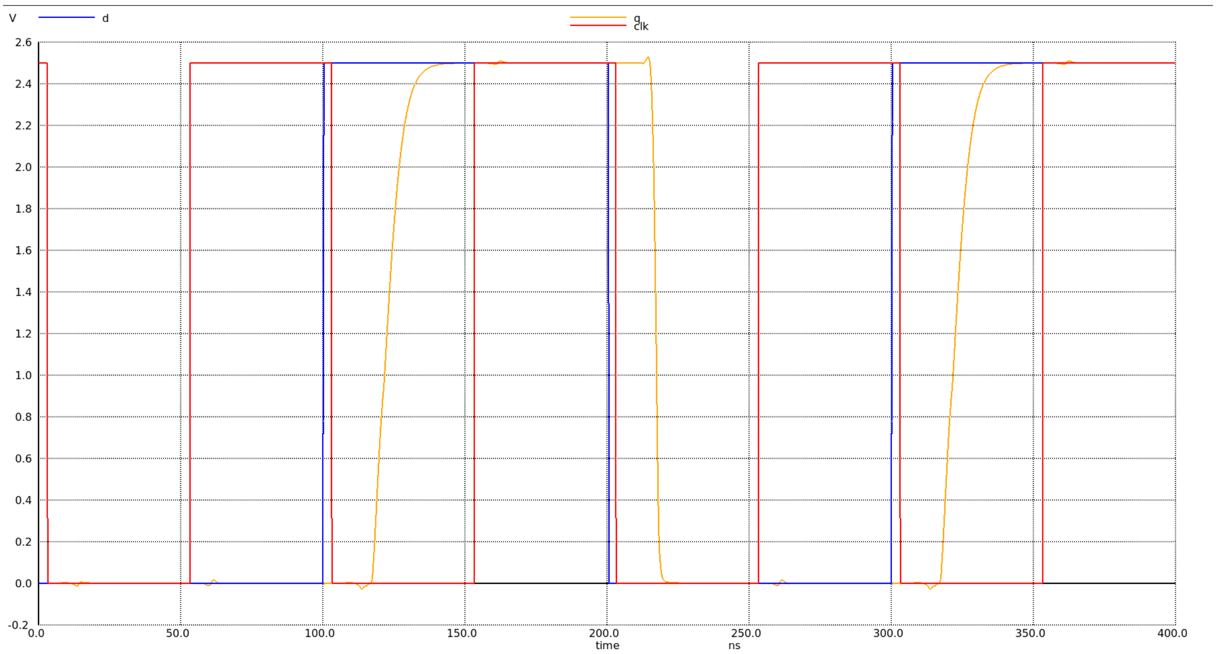

Figure 4: Transient response waveforms of the master-slave register

2.2 (b)

For an input rise/fall time ($t_{rf}(D)$) of 200ps and an assumed output load capacitance (C_Q) of 0.2pF at node Q, the simulated transient response waveforms are illustrated below:

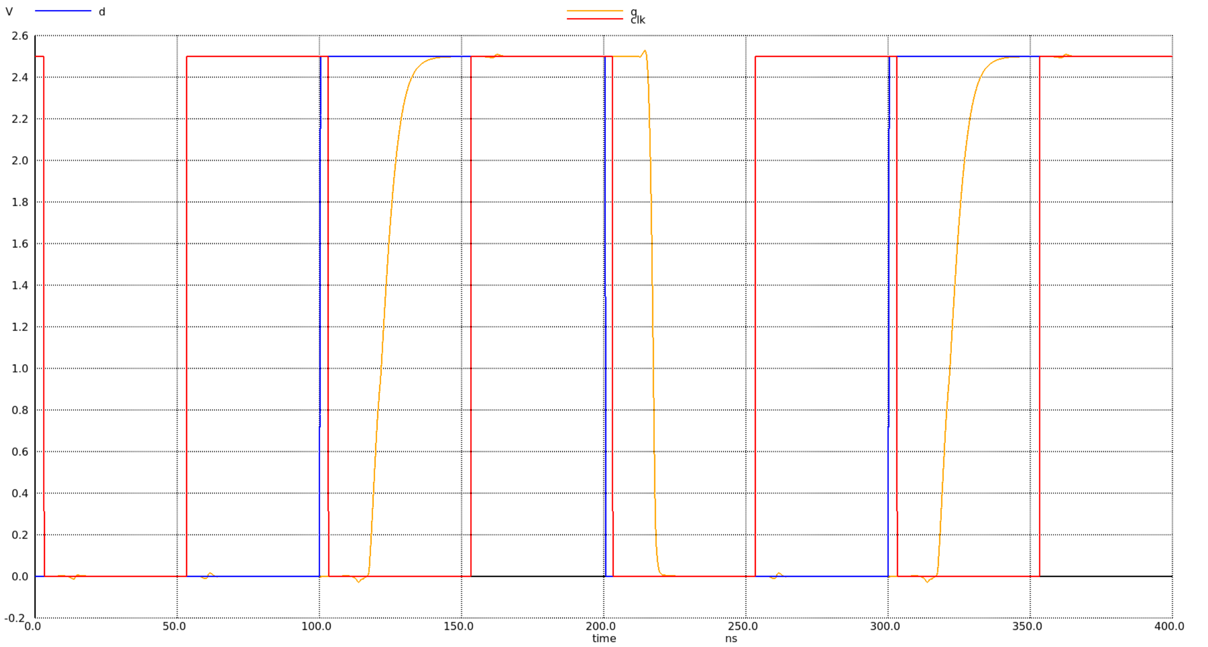

Figure 5: Transient response waveforms of the master-slave register

The measured clock-to-output propagation delays are summarized in Table 3:

Table 3: Measured $CLK \rightarrow Q$ Propagation Delays

Timing Parameter	**Value (ns)**
T_{PHL}	14
T_{PLH}	18
T_{pd} **(Average)**	**16**

To evaluate the rise and fall propagation delays at the intermediate node Q_M, the ngspice simulation tool was utilized. The resulting transient response waveforms are illustrated in Figure 6.

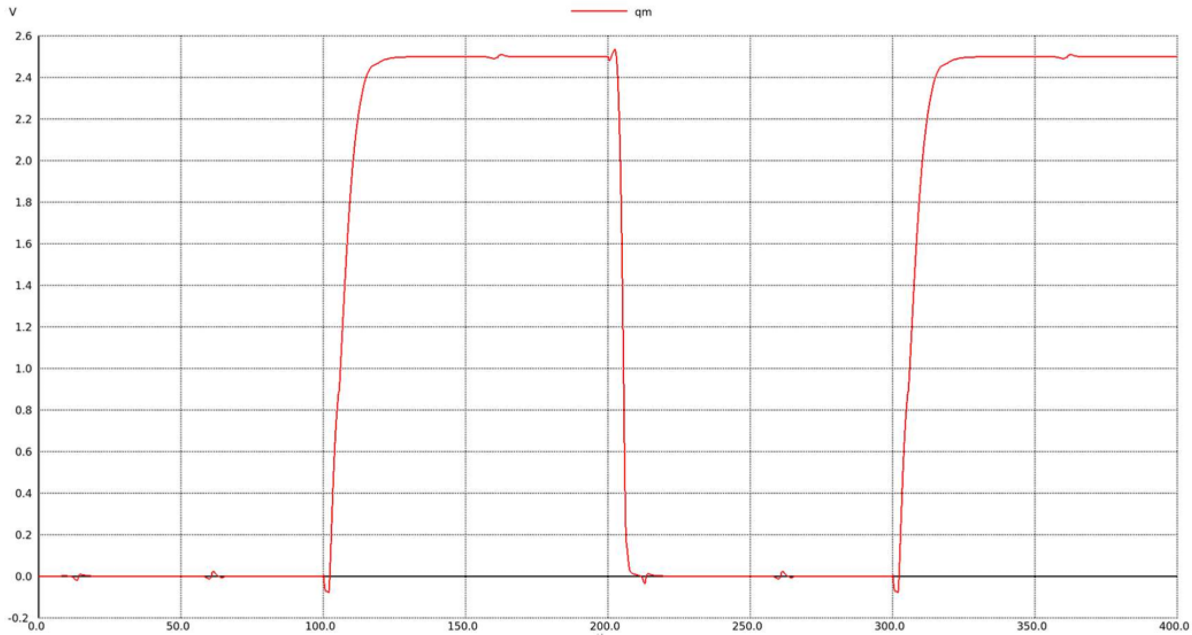

Figure 6: Q_M transient response.

Table 4: Measured Rise/Fall Times and Timing Constraints

Node / Parameter	**Metric**	**Value (ns)**
2*Intermediate Node (Q_M)	Rise Time (t_r)	9.71
	Fall Time (t_f)	1.86
2*Primary Output (Q)	Rise Time (t_r)	7.92
	Fall Time (t_f)	1.89
Setup Time Constraint	T_{setup}	15.00
Hold Time Constraint	T_{hold}	15.30

To further evaluate the sequential and dynamic performance, the system parameters were adjusted according to the assignment requirements. Specifically, for an increased input data rise/fall time ($t_{rf}(D)$) of $400\,\text{ps}$ and a doubled output load capacitance ($C_Q = 2 \times 0.1\,\text{pF} = 0.2\,\text{pF}$) at node Q, the timing characteristics were extracted in an identical manner. The resulting timing metrics and architectural constraints are compiled in Table 5.

Table 5: Measured Characterization Metrics for $t_{rf}(D) = 400\,\text{ps}$ and $C_Q = 0.2\,\text{pF}$

Timing Parameter	**Value (ns)**
$CLK \rightarrow Q$ Propagation Delay (T_{pd})	17.00
Rise Time at Intermediate Node ($t_r(Q_M)$)	9.72
Fall Time at Intermediate Node ($t_f(Q_M)$)	2.35
Rise Time at Primary Output ($t_r(Q)$)	10.96
Fall Time at Primary Output ($t_f(Q)$)	2.25
Setup Time Constraint (T_{setup})	2.90
Hold Time Constraint (T_{hold})	97.50